

NSCOM03 Midterm Recall Drill — Modules 1–4

Write each blanked equation and key term from memory · answers at the back

L01 Review of the Physical & Data Link Layers

L01 · Layers & data link

- OSI bottom-up: (1) _____
- Slide TCP/IP (4): Application = OSI 5–7 · Transport = 4 · (2) _____ · (3) _____
- Slide 2 examples: switch, bridge, Ethernet, PPP → L2; hubs, repeaters, coax, fiber → L1; IP → L3; TCP/UDP → L4
- DLL: frames moved (4) _____ over the physical layer
- DLL services: (5) _____, frame sequencing, flow control, error detection (sometimes correction), (6) _____
- Physical layer: frames → electrical/optical/electromagnetic signals; cables, connectors, pins

L01 · Networks & signals

- LAN typically within (7) _____, copper/fiber, IEEE 802 · WAN: fiber, submarine cable, satellite & terrestrial microwave
- (8) _____ = unused fiber · Starlink (9) _____, 775 satellites (Oct 2020)
- Wireless: unguided, cells; Bluetooth, 802.11, LTE
- IEEE 802: .2 LLC · (10) _____ · .4 token bus · .5 token ring · .11 wireless · .13 none · .16 WLL
- MAC styles: (11) _____ vs (12) _____
- Analog: vary amplitude/frequency/phase · Digital: e.g. 1 = +5 V, 0 = 0 V
- Analog transmission when (13) _____ → modulator / demodulator

L02 Signals, Bandwidth, Noise & Data-Rate Limits

Frequency and period

L02 p6; L02 p7

(14) _____

f frequency — cycles per second (Hz)

T period — the time one cycle takes (s)

Switch between how often a signal repeats and how long one cycle lasts. The prefixes flip: ms $\leftrightarrow$ kHz, μ s $\leftrightarrow$ MHz.

Wavelength

L02 p9

(15) _____

λ wavelength — distance covered during one period (m)

c propagation speed (3×10^8 m/s for light in free space) (m/s)

f frequency (Hz)

T period (s)

How far the signal travels while it completes one cycle — it depends on the medium (through c) and on the frequency.

Bandwidth

L02 p13

(16) _____

B bandwidth — width of the range of frequencies (Hz)

f_{high} highest frequency in the composite signal (Hz)

f_{low} lowest frequency in the composite signal (Hz)

The span of frequencies a composite signal contains (or a medium passes) — not the highest frequency.

Bits per level

L02 p14

(17) _____

L number of signal levels

A digital signal with L levels carries $\log_2 L$ bits in each level: 2 levels $\rightarrow$ 1 bit, 4 $\rightarrow$ 2, 8 $\rightarrow$ 3.

Decibel

L02 p21; L02 p22

(18) _____

dB gain (+) or loss (–) between two points (dB)

P_1 power at the first point (W)

P_2 power at the second point (W)

Compare two powers. Negative = attenuated, positive = amplified; the dB values of a chain of stages simply add.

Signal-to-noise ratio

L02 p25

(19) _____

SNR signal-to-noise ratio, a plain ratio

P_{signal} average signal power (W)

P_{noise} average noise power (W)

SNR_{dB} the same ratio in decibels (dB)

How far the signal stands above the noise. Shannon needs the plain ratio — convert from dB first.

Nyquist bit rate (noiseless)

L02 p29

(20) _____

N maximum bit rate of a noiseless channel (bps)

B channel bandwidth (Hz)

L number of signal levels

Noiseless channel: the most bits per second a bandwidth B can carry with L levels. More levels raise N , but the receiver must still tell them apart.

Shannon capacity (noisy)

L02 p31

(21) _____

C capacity — the upper limit on the bit rate (bps)

B channel bandwidth (Hz)

SNR signal-to-noise ratio as a plain ratio (not dB)

Noisy channel: the highest possible bit rate, whatever the signalling. There is no L in it — use Shannon for the limit, then Nyquist to choose L .

Latency

L02 p35

(22) _____

T_{prop} propagation time = distance ÷ propagation speed (s)

T_{trans} transmission time = message size (bits) ÷ bandwidth (bps) (s)

T_{queue} waiting time in device queues (s)

T_{proc} processing time in devices (s)

Total time for a whole message to arrive. Long distances make propagation dominate; large messages on slow links make transmission dominate.

L02 · Signals & bandwidth

- Analog = infinite levels · digital = limited levels; periodic vs nonperiodic
- (23) _____ (Hz) · 4 cycles in 1 s → 4 Hz, $T = \frac{1}{4}$ s · phase $0^\circ/90^\circ/180^\circ/270^\circ = 0\uparrow$ / peak / $0\downarrow$ / trough
- (24) _____, $c = 3 \times 10^8$ m/s · red light 4×10^{14} Hz → (25) _____
- Composite = sum of sines (Fourier); (26) _____ (5000 – 1000 = 4000 Hz)
- Digital signal = composite with (27) _____; L levels → $\log_2 L$ bits per level
- Baseband → low-pass channel (first harmonic $B = N/2$) · broadband → modulation onto a (28) _____ channel

L02 · dB, SNR & noise

- (29) _____; loss negative, gain positive; half power = (30) _____
- dB of cascaded stages add; $P_{\text{out}} = P_{\text{in}} \times 10^{(\text{dB}/10)}$
- (31) _____; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \cdot 10 \text{ mW} / 1 \mu\text{W} \rightarrow 10,000 \rightarrow$ (32) _____
- Noise: (33) _____ (electrons), (34) _____ (motors), (35) _____ (adjacent wire), (36) _____ (spikes)
- Attenuation = energy loss · distortion = shape change

L02 · Rate limits & latency

- Nyquist (noiseless): (37) _____ · 3000 Hz, $L = 2 \rightarrow$ (38) _____
- Shannon (noisy): (39) _____, SNR linear · 3000 Hz, 3162 → (40) _____
- Slide 33: 1 MHz, SNR 63 → $C =$ (41) _____; use 4 Mbps → (42) _____
- (43) _____
- $T_{\text{prop}} = d/v$ · $T_{\text{trans}} = \text{size}/\text{bandwidth}$ · 12,000 km at 2.4×10^8 m/s → (44) _____; 2.5 KB at 1 Gbps → (45) _____
- Throughput < bandwidth · jitter = variation in delay

Signal (baud) rate

L03 p8

(46) _____

 S signal rate — signal elements per second (baud) c case factor: worst 1, best 0, average $\frac{1}{2}$ N data rate (bps) r data elements carried by one signal element

How many signal elements per second a line code needs for data rate N . Use the average case $c = \frac{1}{2}$ unless the question says otherwise.

Minimum bandwidth

L03 p25; L03 p46

(47) _____

 $B_{\min}$ minimum bandwidth the line code needs (Hz) S_{avg} average signal rate (baud)

Per scheme (slide 46): NRZ-L, NRZ-I, AMI $\rightarrow N/2 \cdot RZ$, Manchester, differential Manchester $\rightarrow N \cdot 2B1Q \rightarrow N/4 \cdot MLT-3 \rightarrow N/3 \cdot 8B6T \rightarrow 3N/4 \cdot 4D-PAM5 \rightarrow N/8$.

mBnL code condition

L03 p36; L03 p37

(48) _____

 m data bits in a block (data elements) n signal elements per block L levels per signal element: $B = 2$, $T = 3$, $Q = 4$

An mBnL code works only if the L^n signal patterns are at least as many as the 2^m data patterns; the patterns left over are redundancy.

4B/5B rate

L03 p54

(49) _____

 N_{line} bit rate on the line after 4B/5B (bps) N data rate before block coding (bps)

Every 4 data bits become 5 code bits, so the line carries 25% more bits; then apply the line code to the new rate.

L03a · Rates & properties

- (50) _____ (baud); $c = \frac{1}{2}$ average; $B_{\min} = S$
- r : NRZ 1 · RZ/Manchester $\frac{1}{2}$ · (51) _____ · 8B6T $\frac{4}{3}$
- Avg B: NRZ-L/I, AMI (52) _____ · RZ, biphase (53) _____ · 2B1Q $N/4$ · MLT-3 $N/3$ · 8B6T $3N/4$ · 4D-PAM5 $N/8$
- Good code: no (54) _____, no (55) _____, self-sync, error detection, noise immunity, low complexity
- Ex 1: 100 kbps, $r = 1 \rightarrow$ (56) _____ · Ex 2: 0.1% fast $\rightarrow$ 1001 bps / 1,001,000 bps · Ex 3: NRZ-I 1 Mbps $\rightarrow$ (57) _____

L03a · Scheme rules (slide figures)

- Unipolar NRZ 1 = +V, 0 = 0 · (58) _____ · (59) _____ (start +)
- RZ: 1 = +V $\rightarrow$ 0, 0 = -V $\rightarrow$ 0 · (60) _____
- (61) _____, always mid-bit transition
- (62) _____ (first +) · pseudoternary = reverse (low priority)
- 2B1Q (prev +): 00 +1, 01 +3, 10 -1, 11 -3 (prev - : signs flip)
- MLT-3: 0 stays; 1 $\rightarrow$ 0 from $\pm V$, or $\rightarrow$ opposite of last nonzero from 0
- mBnL needs (63) _____; $B = 2$, $T = 3$, $Q = 4$ levels

L03a · Block coding & scrambling

- Block coding mB/nB: division → substitution → combination; adds redundancy
- (64) _____; rate $\times 5/4$; 16 spare codes (control: I 11111, Q 00000, H 00100, J 11000, K 10001, T 01101)
- Ex 4: 1 Mbps → (65) _____ → NRZ-I (66) _____ / Manchester 1.25 MHz
- $8B/10B = 5B/6B + 3B/4B +$ disparity controller
- (67) _____ (V same polarity as previous pulse)
- (68) _____ (pulses since last substitution)

Sampling rate

L03 p64

(69) _____

 f_s sampling rate (samples/s) T_s sampling interval (s)

Samples per second from the time between samples, and back.

Nyquist sampling theorem

L03 p66; L03 p67

(70) _____

 f_s sampling rate (samples/s) $f_{\max}$ highest frequency in the signal (Hz)

Sample at least twice the highest frequency — for low-pass and bandpass signals alike. A bandpass signal's bandwidth alone is not enough: you need its highest frequency.

Zone height (quantization step)

L03 p72

(71) _____

 Δ zone height (quantization step) (V) $V_{\max}$ top of the amplitude range (V) $V_{\min}$ bottom of the amplitude range (V) L number of zones (levels)

Split the amplitude range into L equal zones; each sample is replaced by the midpoint of its zone.

Bits per sample

L03 p75

(72) _____

 n_b bits per sample L number of zones (levels)

The number of bits needed to give every zone its own code.

PCM bit rate

L03 p79

(73) _____

 N PCM bit rate (bps) n_b bits per sample f_s sampling rate (samples/s)

The bit rate PCM produces: bits per sample $\times$ samples per second.

Minimum PCM bandwidth

L03 p83

(74) _____

 $B_{\min}$ minimum bandwidth of the PCM signal (Hz) n_b bits per sample $f_{\max}$ highest frequency of the analog signal (Hz) N PCM bit rate (bps)

Minimum bandwidth when the PCM bits are sent with an NRZ-type line code ($c = 1/2$, $r = 1$).

Slide example 10: $8 \times 4 \text{ kHz} = 32 \text{ kHz}$.

Quantization error bound

L03 p76; L03 p78

(75) _____

 Δ zone height (V)

The midpoint of a zone is never more than half a zone away from the true sample.

L03b · PCM

- PCM = (76) _____; filter first
- $f_s = 1/T_s$; Nyquist (77) _____ (low-pass and bandpass); voice 4 kHz → (78) _____
- Bandpass with only B known → rate (79) _____
- Sampling: ideal (impulse) · natural (follows curve) · (80) _____ (sample and hold) → PAM
- (81) _____; zones 0 ... L-1 from the bottom; sample → (82) _____; (83) _____
- -20 ... +20 V, L = 8 → $\Delta = 5$ V; 000 = -20 to -15 V ... 111 = +15 to +20 V
- $|\text{error}| \leq \Delta/2$; low amplitudes suffer → non-linear quantization, (84) _____

L03b · Rates, DM, modes

- (85) _____ — voice, 8 bits → (86) _____
- (87) _____ (NRZ-type) — 8×4 kHz = (88) _____
- Decoder: (89) _____ + low-pass filter (cutoff = $f_{\max}$) = quantized difference
- DM: 1 bit per sample, 1 = up δ , 0 = down δ ; DPCM = quantized difference
- Parallel: n bits per tick on n wires · Serial: 1 bit per tick
- Async: (90) _____ + byte + (91) _____, gaps OK · Sync: no start/stop, frames · Isochronous: equal gaps

Baud (signal) rate

L04 p5

(92) _____

 S baud rate — signal elements per second (baud) N bit rate (bps) r data bits carried by one signal element

Analog transmission of digital data: each signal element carries r bits, so $S \leq N$.

Analog transmission of digital data: $S \leq N$.

Bits per element $\leftrightarrow$ levels

L04 p7

(93) _____

 r bits per signal element L number of distinct signal elements (levels, phases, frequencies)

Bits per signal element from the number of distinct elements, and back.

ASK / PSK bandwidth

L04 p9; L04 p21

(94) _____

 B bandwidth (Hz) d modulation and filtering factor, $0 \leq d \leq 1$ S baud rate (baud)

Bandwidth of ASK and of PSK. With $d = 0$ the bandwidth is at its minimum, $B = S$.

Binary FSK bandwidth

L04 p14; L04 p15

(95) _____

 B bandwidth (Hz) d modulation and filtering factor, $0 \leq d \leq 1$ S baud rate (baud)

Δf half the carrier spacing: bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$ (Hz)

Binary FSK needs the ASK bandwidth plus the spacing $2\Delta f$ between its two carriers.

Multilevel FSK bandwidth

L04 p18; L04 p19

(96) _____

 B bandwidth (Hz) S baud rate (baud) L number of frequencies

Δf half the spacing between neighbouring frequencies (Hz)

L frequencies, each $2\Delta f$ apart. With $d = 0$ and the minimum spacing $2\Delta f = S$, this becomes $B = L \times S$.

$\triangle$ Slide 18 prints " $(L - 1)/2\Delta f$ " — it is a product.

Carrier in an available band

L04 p12; L04 p13

(97) _____

 f_c carrier frequency (Hz) f_{low} bottom of the available band (Hz) f_{high} top of the available band (Hz)

Put the carrier in the middle of the band. Full duplex: split the band into two halves with a carrier in the middle of each.

L04 · Digital → analog

- **ASK** amplitude · **FSK** frequency · **PSK** phase · **QAM** amplitude + phase
- (98) _____ (baud); (99) _____, $L = 2^f$; $S \leq N$
- ASK/PSK: (100) _____, $0 \leq d \leq 1$; BASK = OOK, $r = 1$
- FSK: (101) _____; bit 1 → $f_c + \Delta f$, bit 0 → $f_c - \Delta f$
- MFSK: (102) _____ = $L \cdot S$ when $d = 0$, $2\Delta f = S$
- Carrier at band middle; full duplex → halve the band (200–300 kHz → 225 and 275 kHz)
- BASK = (103) _____ NRZ × carrier; BPSK = (104) _____ NRZ × carrier; FSK via VCO
- PSK more robust than ASK (noise hits amplitude)

L04 · Constellations & examples

- $x = I$ (in-phase), $y = Q$ (quadrature); length = amplitude, angle = phase
- OOK: 0 at origin, 1 on +I · BPSK: 0 at -I, 1 at +I
- QPSK (1st bit → I): (105) _____
- 16-QAM (slide 31): first 2 bits = quadrant (00 I, 10 II, 11 III, 01 IV); 1100 → 225°
- $r = 4$, $S = 1000$ → (106) _____ · $N = 8000$, $S = 1000$ → (107) _____
- ASK 200–300 kHz, $d = 1$ → (108) _____ · FSK, $2\Delta f = 50$ kHz → (109) _____
- QPSK 12 Mbps, $d = 0$ → (110) _____ · MFSK 3 bits, 3 Mbps → (111) _____

Σ Exam conventions, mental math and traps

Symbol clashes

- **S** = (112) _____ in baud: $S = c \cdot N/r$ (L03), $S = N/r$ (L04). **C** = (113) _____ in bps. Not the same thing.
- **c** = (114) _____ (0 to 1, average $\frac{1}{2}$) in $S = c \cdot N/r$. In L02, **c** = (115) _____, 3×10^8 m/s.
- **N** = (116) _____ in bps. **N_{max}** = (117) _____ is Nyquist's limit for a noiseless channel.

Mental-math tips (no calculator)

- $\log_2$ of a power of 2 = (118) _____ ($\log_2 64 = 6$); $\log_2 1000 \approx$ (119) _____
- dB $\rightarrow$ ratio: (120) _____; ratio $\rightarrow$ dB: $10 \cdot \log_{10}(\text{ratio})$. Add dB to multiply ratios.
- (121) _____ $\cdot$ (122) _____ $\cdot 20$ dB = $\times 100 \cdot -3$ dB $\approx \times \frac{1}{2}$
- (123) _____ (1024 vs 1000)
- Powers of 2: $2^1=2 \cdot 2^2=4 \cdot 2^3=8 \cdot 2^4=16 \cdot 2^5=$ (124) _____ $\cdot 2^6=(125)$ _____ $\cdot 2^7=(126)$ _____ $\cdot 2^8=(127)$ _____ $\cdot 2^9=(128)$ _____ $\cdot 2^{10}=(129)$ _____
- $\text{SNR} = 2^k - 1$ gives $\log_2(1 + \text{SNR}) = k$: SNR 3, 7, 15, 31, 63, 255 $\rightarrow$ 2, 3, 4, 5, 6, 8

Unit traps

- (130) _____ for data rates and sizes (1 kbps = 1000 bps; 1 KB = 1000 B on the slides), not 1024.
- Bytes (131) _____ = bits (2.5 KB = 20,000 bits) before dividing by a bit rate.
- (132) _____, (133) _____: 0.020 ms = 20 μ s.
- Convert everything to base units (Hz, bps, s, m) before substituting.

Answers — recall drill

Cover this page until every blank is filled in from memory. 133 answers, numbered as on the sheet.

- (1) Physical, Data link, Network, Transport, Session, Presentation, Application
- (2) Internet = 3
- (3) Network access = 1–2
- (4) error-free between two end nodes
- (5) framing
- (6) QoS
- (7) 5 km
- (8) Dark fiber
- (9) 550 km
- (10) .3 Ethernet (CSMA/CD)
- (11) ALOHA style
- (12) token style
- (13) bandwidth is limited
- (14) $f = \frac{1}{T}$ $T = \frac{1}{f}$
- (15) $\lambda = \frac{c}{f} = c \times T$
- (16) $B = f_{\text{high}} - f_{\text{low}}$
- (17) bits per level = $\log_2 L$
- (18) $\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$
- (19) $\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$ $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$
- (20) $N = 2 \times B \times \log_2 L$
- (21) $C = B \times \log_2(1 + \text{SNR})$
- (22) $\text{Latency} = T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$
- (23) $f = 1/T$
- (24) $\lambda = c/f$
- (25) 0.75 μm
- (26) $B = f_{\text{high}} - f_{\text{low}}$
- (27) infinite bandwidth
- (28) bandpass
- (29) $\text{dB} = 10 \log_{10}(P_2/P_1)$
- (30) –3 dB
- (31) $\text{SNR} = P_s/P_n$
- (32) 40 dB
- (33) thermal
- (34) induced
- (35) crosstalk
- (36) impulse
- (37) $N = 2B \log_2 L$
- (38) 6000 bps
- (39) $C = B \log_2(1 + \text{SNR})$
- (40) ≈ 34.86 kbps
- (41) 6 Mbps
- (42) $L = 4$
- (43) $\text{Latency} = T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$
- (44) 50 ms
- (45) 0.020 ms
- (46) $S = c \times N \times \frac{1}{r}$
- (47) $B_{\text{min}} = S_{\text{avg}}$
- (48) $2^m \leq L^n$
- (49) $N_{\text{line}} = N \times \frac{5}{4}$
- (50) $S = c \times N \times 1/r$
- (51) 2B1Q 2
- (52) $N/2$
- (53) N
- (54) baseline wandering
- (55) DC
- (56) 50 kbaud
- (57) 500 kbaud, 500 kHz
- (58) NRZ-L 0 = +V, 1 = –V
- (59) NRZ-I: 1 inverts
- (60) Manchester 0 = high→low, 1 = low→high
- (61) Diff. Manchester: 0 = transition at start
- (62) AMI 0 = 0 V, 1s alternate
- (63) $2^m \leq L^n$
- (64) 4B/5B + NRZ-I
- (65) 1.25 Mbps
- (66) 625 kHz
- (67) B8ZS: 8 zeros → 000VB0VB
- (68) HDB3: 4 zeros → B00V if even, 000V if odd
- (69) $f_s = \frac{1}{T_s}$
- (70) $f_s \geq 2 \times f_{\text{max}}$

$$(71) \Delta = \frac{V_{\max} - V_{\min}}{L}$$

$$(72) n_b = \log_2 L$$

$$(73) N = n_b \times f_s$$

$$(74) B_{\min} = n_b \times f_{\max} = \frac{N}{2}$$

$$(75) |\text{error}| \leq \frac{\Delta}{2}$$

(76) sample → quantize → encode

$$(77) f_s \geq 2 \cdot f_{\max}$$

(78) 8000 sps

(79) cannot be determined

(80) flat-top

$$(81) \Delta = (\max - \min)/L$$

(82) midpoint

$$(83) n_b = \log_2 L$$

(84) companding

$$(85) N = n_b \times f_s$$

(86) 64 kbps

$$(87) B_{\min} = n_b \times f_{\max}$$

(88) 32 kHz

(89) hold circuit

(90) start 0

(91) stop 1

$$(92) S = N \times \frac{1}{r}$$

$$(93) r = \log_2 L \Leftrightarrow L = 2^r$$

$$(94) B = (1 + d) \times S$$

$$(95) B = (1 + d) \times S + 2\Delta f$$

$$(96) B = (1 + d) \times S + (L - 1) \times 2\Delta f$$

$$(97) f_c = \frac{f_{\text{low}} + f_{\text{high}}}{2}$$

$$(98) S = N/r$$

$$(99) r = \log_2 L$$

$$(100) B = (1 + d)S$$

$$(101) B = (1 + d)S + 2\Delta f$$

$$(102) B = (1 + d)S + (L - 1) \cdot 2\Delta f$$

(103) unipolar

(104) polar

$$(105) 11 \rightarrow 45^\circ, 01 \rightarrow 135^\circ, 00 \rightarrow -135^\circ, 10 \rightarrow -45^\circ$$

(106) 4000 bps

$$(107) r = 8, L = 256$$

(108) 250 kHz, 50 kbps

(109) 25 kbps

(110) 6 MHz

(111) L = 8, 1 Mbaud, 8 MHz

(112) signal (baud) rate

(113) Shannon capacity

(114) case factor

(115) speed of light

(116) bit (data) rate

$$(117) 2 \cdot B \cdot \log_2 L$$

(118) count the doublings

(119) 10

$$(120) 10^{(\text{dB}/10)}$$

$$(121) 3 \text{ dB} \approx \times 2$$

$$(122) 10 \text{ dB} = \times 10$$

$$(123) 2^{10} \approx 10^3$$

(124) 32

(125) 64

(126) 128

(127) 256

(128) 512

(129) 1024

(130) K = 1000

(131) $\times 8$

$$(132) \mu\text{s} = 10^{-6} \text{ s}$$

$$(133) \text{ms} = 10^{-3} \text{ s}$$

END OF DOCUMENT: recall · build r1